

A Lean 4 formalization of “ $\zeta(5)$ is irrational” (A. Fauzan, 17 September 2026)

Summary of <https://github.com/danromik/zeta5-irrationality>

24 September 2026

Abstract

The repository contains a complete formalization in Lean 4, with Mathlib, of the proof in A. Fauzan’s preprint “ $\zeta(5)$ is irrational”. The main theorem, `Zeta5.zeta5_irrational : Irrational zeta5`, is proved with no `sorry`. It depends on the three standard axioms of Lean and on one external axiom, the prime number theorem in the form $\theta(x) \sim x$, which Mathlib does not contain. The formal proof passes through the paper’s intermediate statements (Theorem 2.1, Propositions 2.2, 4.1, 4.3, 5.1, 5.2, 6.3, Lemma 3.3 and the numbered equations between them); at the places listed in Section 4 it proves a step by a different argument from the paper’s.

1 The theorem and its assumptions

```
def zeta5 : ℝ := ∑' v : ℕ, (1 : ℝ) / ((v : ℝ) + 1) ^ 5
theorem zeta5_irrational : Irrational zeta5
```

Both are in the namespace `Zeta5`; `Irrational` is Mathlib’s predicate. `#print axioms Zeta5.zeta5_irrational` gives

```
[propext, Classical.choice, Quot.sound, Zeta5.Axioms.chebyshev_theta_asymptotic]
```

The last is the only axiom declared in the project (`Zeta5/Axioms.lean`):

```
axiom chebyshev_theta_asymptotic :
  Tendsto (fun x : ℝ => Chebyshev.theta x / x) atTop (nhds 1)
```

where `Chebyshev.theta` is Mathlib’s $\theta(x) = \sum_{p \leq x} \log p$. This is the prime number theorem (Hadamard, de la Vallée Poussin, 1896) as stated in textbooks, e.g. Apostol, *Introduction to Analytic Number Theory*, Thm. 4.4 with Ch. 13; the paper cites DLMF 27.12.2–27.12.4. It is used only in Proposition 5.2, through the partial-summation theorem `Zeta5.PNT.prime_riemann_sum`. Lean therefore checks the implication

$$\text{prime number theorem} \implies \zeta(5) \notin \mathbb{Q}.$$

2 The build

Lean 4.34.0, Mathlib v4.34.0; 70 files under `Zeta5/`, about 29 400 lines. `lake build` ends with `Build completed successfully (8995 jobs)`, with no error and no `sorry` warning; the 17 warnings are linter warnings (unused section variables and unreferenced hypothesis names). A clean build takes about 7 minutes on a 12-core machine.

3 Structure

Paper	Lean files (in Zeta5/)
§2, Proposition 2.2	<code>Functional</code> , <code>Positivity</code> ; the pole integral of p. 5 in <code>Hermite</code>
§3, Lemma 3.3, (3.11)–(3.12)	<code>LocalFunctional</code> , <code>Lemma33</code> , <code>CrudeBound</code> , <code>Arithmetic</code>
§4.1, Proposition 4.1	<code>Counting</code> , <code>Section41</code> , <code>Section3</code> , <code>InnerTate</code> , <code>InnerGeneral</code> , <code>InnerEntries</code> , <code>HermiteBasis</code>
§4.2, Proposition 4.3	<code>Lemma42</code> , <code>OuterLocal</code> , <code>OuterBasis</code> , <code>OuterRange</code>
§5, Propositions 5.1, 5.2	<code>Normalization</code> , <code>Uniformity</code> , <code>PNT</code> , <code>PrimeSum</code> , <code>Tail</code>
Appendix B	<code>AppendixB</code>
§6, Appendix A, Proposition 6.3	<code>RealBound</code> , <code>Sec6/*</code> (37 files)
§7, Theorems 2.1, 1.1	<code>Asymptotics</code> , <code>Skeleton</code> , <code>Interface</code>

`Interface.lean` contains, for each numbered statement of the paper used by Theorem 1.1, a declaration whose type transcribes the printed statement. The finite computations of the paper (Appendix B’s 143 intervals and Tables 3–4, Table 1 and the constants of Appendix A.4, the margin (7.2)) and the certified partition used for (6.2) are proved by exact rational arithmetic and certified enclosures; the project contains no `native_decide`, `implemented_by` or `extern`.

4 Deviations from the paper

Each item is recorded in the docstring of the declaration concerned.

1. **Lemma 3.2, (3.7).** The Tate algebra, (3.5) and (3.6) are not formalized; the valuation bound that Proposition 4.1 needs is proved from Raabe’s multiplication theorem for τ and a truncated inverse of the far-pole factor.
2. **Lemma 3.3, polynomial part.** Proved through the binomial basis $\binom{x+K}{k}$ and an exact computation of τ , in place of (3.8)–(3.9). The term $-v_p(24)$ of (3.10) is unnecessary.
3. **Unimodularity of (4.5) and (4.11),** from one lemma on bases that reduce mod p to a Hermite-interpolation basis.
4. **The inequality $b_a \leq 6\alpha x + 3$ (p. 10),** stated in the paper without proof, is proved in the strict form $p\ell_N(a) < 2N + p$.
5. **The divided difference of p. 12** uses the term $-1/4 + 1/(2j)$ of (2.3) in addition to the cited congruence.
6. **(4.10):** the vanishing is proved in the monomial basis, and $\text{rank } L \leq r_p$ is transported to the basis (4.11).
7. **(5.16)–(5.17)** are proved in the combined form that (5.18) uses.
8. **The pole integral of p. 5** is proved from the series for w and the Mittag-Leffler expansion of the cotangent, not from Hermite’s formula.

9. **Partial summation (p. 15)** is proved from $\theta(x) \sim x$ by a Darboux-type sandwich.
10. **(6.6)–(6.9)**. The kernel is regularized on the real line, $\frac{1}{2} \log(x^2 + \varepsilon^2)$ with $\varepsilon = (16K)^{-2}$, in place of circles in $\mathbb{C}$; Lemma 6.2 is proved for this kernel from a Frullani representation and the positive definiteness of the Gaussian kernel. The smoothing error is bounded directly. The configuration bound obtained has $20h$ in place of $(120 + \sqrt{2})h$.
11. **(A.2)** is used as the inequality $I(\rho) \leq I_k(\rho, \rho)$ for the regularized kernel.
12. **(6.10)–(6.13)**. The factor $1/h!$ is kept, and $326K^6$ replaces 652.
13. **(6.2) on $[0, 2]$** is proved by kernel-checked interval arithmetic on the closed forms (A.1), (A.5) over a partition of 1049 cells, in place of (A.9) on Table 2; for $t \geq 2$, two cells cover $[2, 4]$ and an analytic bound covers $t \geq 4$.

5 Verification

To accept the result, a reader needs to trust Lean’s kernel and to read two statements, both written with Mathlib names only in `cert/Check.lean`:

```
example : Irrational ( $\sum' v : \mathbb{N}, (1 : \mathbb{R}) / ((v : \mathbb{R}) + 1) ^ 5$ ) :=
  Zeta5.zeta5_irrational
example : Tendsto (fun x :  $\mathbb{R} \Rightarrow$  Chebyshev.theta x / x) atTop (nhds 1) :=
  Zeta5.Axioms.chebyshev_theta_asymptotic
```

together with the Mathlib definitions of `Irrational`, `tsum` and `Chebyshev.theta`. The checks are `lake build, lake env lean cert/Check.lean` (which prints the axiom list of Section 1), and `lake env leanchecker Zeta5`, which re-checks every declaration of the compiled modules with the kernel alone (`docs/CERTIFICATION.md`).